

MUHAMMAD DANIYAL

📞 92+3042327945 ✉ daniyalamir429@gmail.com 🔗 linkedin.com/in/daniyal_khan

SUMMARY — Aspiring Undergraduate of BE Metallurgy and Materials Engineering. Equipped with a strong foundation in materials science and metallurgical processes. Experienced in laboratory research and analysis, with hands-on proficiency in material characterization techniques, corrosion control and failure analysis. Eager to contribute to a forward-thinking organization and grow professionally in the field of metallurgy and materials engineering.

TECHNICAL COMPETENCIES

Material Characterization	XRD, DLS	Corrosion Testing	Electrochemical methods and Corrosion prevention techniques.
Softwares	SolidWorks	Failure Analysis	Analysis of material failures using Fractography, Metallography and Non-Destructive Testing (NDT)
Metallurgical Process	Heat Treatment, Casting and Forming Techniques, Powder Metallurgy		
Mechanical Testing	Tensile, Compression, Hardness, Creep, Fatigue and Charpy Impact Testing		

EDUCATION

Mehran University of Engineering and Technology <i>Bachelor of Engineering in Metallurgy and Materials</i>	Jan 2021-25
Govt Faqir Ghulam Rasool H/SEC School Jamshoro <i>Pre Engineering</i>	July 2018-20
GSC O'Levels <i>Science</i>	March 2016-18

RECENT EXPERIENCE

Alsons Group of Industries Ltds <i>Trainee Engineer</i>	Dec 2024- Present
Agha Steel Industries Ltds <i>Metallurgy and Materials Engineer, Intern</i> - Assisted senior engineer in conducting metallurgical testing and analysis. - Contributed to quality control processes and inspection. - Participated in heat treatment experiments and metallographic studies - Prepared technical report summarizing findings and recommendation	June 2022 to July. 2022
SSGC Sui Southern Gas Company Limited <i>Corrosion Control Intern</i> - Participated in field inspection to measure pipe to soil potential range ensuring compliance with industry standards. - Evaluated and recorded cathodic test points to monitor and maintain pipeline integrity. - Explore different strategies for mitigating corrosion, such as cathodic protection, coatings, and inhibitors, and understand their application in real-world scenarios. - Developed technical report summarising findings and suggesting improvements for pipeline corrosion prevention.	3 Jan 2024 to 31 Jan 2024

PROJECTS

- Synthesis And Characterization of Phase Change Material (PCM) for Cooling Applications (FINAL YEAR PROJECT)**
To synthesize a Phase Change Material (PCM) by using paraffin wax and hybrid of Al₂O₃ and TiO₂ and to study the cooling behavior of (PCM)
- Development of color pigments through mineral extraction of iron oxide**
Collected extracted mineral from the mines and physically altered the structure of iron oxide involving grinding, milling, or crushing to achieve the desired particle size and distribution, which affect the pigment's color intensity and dispersion properties.
- Complex Engineering Problem of Foundry Engineering**
Did research on the material and procedure of designing a vertical axis wind turbine using the wax pattern casting of aluminum.

CERTIFICATIONS

- **Two Day short course on Non Destructive Testing**
- **Introduction to Entrepreneurship and Business Model Canvas Workshop**
- **1 point CPD (Continuing Professional Development Program) on "Corrosion Prevention and Control"**

EXTRA CURRICULAR ACTIVITIES/ STARTUPS

- **MEDWE Colors**
Founder of Medwe Colors, a startup focused on developing innovative color pigments.
- **Phase Genius**
Founder of Phase Genius, a startup focused on developing phase changing material for cooling applications.
- **Vice President of Metallurgy and Materials Advantage Chapter(MMAC)**
Coordinated and managed events, workshops, and meetings. Collaborated with society members and university staff to plan and execute initiatives and events .